Transformer Blocks

CSE 256 PA2 – Part A: Encoder and Decoder

Jai Dayanand
UC San Diego | Fall 2026

Experimental Setup (Parts 1–2)

I implemented a Transformer encoder for speaker classification and a causal Transformer decoder for next-word prediction. Attention is computed directly from query, key, and value projections, scaled dot products, masking, and softmax. Neither model uses a built-in Transformer or multihead-attention layer. The pretrained-model analysis is documented separately in the executed Part 3 notebook, which forms Part B.

Setting	**Value**
Vocabulary size	5,634
Embedding dimension	64
Attention heads / Transformer blocks	2 / 4
Maximum sequence length	32 tokens
Feedforward hidden width / activation	100 / ReLU
Optimizer / learning rate	Adam / 0.001
Batch size / seed	16 / 42
Execution environment	CPU; PyTorch 2.15.0+cpu
Classifier training	15 epochs
Decoder training	500 updates; 256,000 target positions

Tokenization and architecture

I used the supplied NLTK word-level tokenization, with sorted vocabulary IDs for repeatability. Following the starter, the vocabulary is built from the two training files only; test files do not contribute vocabulary entries. Padding and unknown words have reserved IDs 0 and 1. Each model adds learned absolute position embeddings to token embeddings before its first block.

Each block uses pre-layer normalization, an attention residual connection, and a ReLU feedforward residual connection. A final layer norm is applied after the last block. The encoder masks padding keys. The decoder applies a causal mask and uses a separate, untied vocabulary projection.

Part 1: Encoder Trained with a Classifier (30 points)

1.1 Encoder. The encoder produces contextual vectors for all input tokens using four custom Transformer blocks and learned absolute position embeddings.

1.2 Feedforward classifier. The classifier maps the mean-pooled 64-dimensional sequence representation through a 100-dimensional ReLU hidden layer to three speaker logits.

1.3 Joint training. The encoder and classifier are optimized together with cross-entropy on the speaker labels.

1.4 Sanity checks. Attention rows were tested to sum to one; the two corresponding attention maps and interpretation appear on the following page.

1.5 Evaluation. Test accuracy is recorded after every one of the 15 epochs, with the final value highlighted below.

The three classes are Obama (0), George W. Bush (1), and George H. W. Bush (2). Sequences are truncated to 32 tokens; padding positions are excluded from mean pooling.

Encoder parameters: 482,192. **Encoder plus classifier:** 488,995.

Epoch	Training loss	Test accuracy (%)
1	1.0608	46.53
2	0.9989	42.80
3	0.9063	59.73
4	0.7778	64.40
5	0.5989	70.13
6	0.4433	76.27
7	0.2742	79.60
8	0.1817	83.47
9	0.1376	83.60
10	0.1167	85.33
11	0.0650	86.13
12	0.0555	85.60
13	0.0631	84.00
14	0.0391	83.73
15	0.0198	83.87

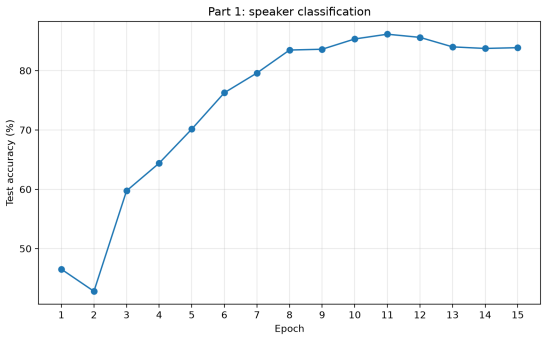


Figure 1: Test accuracy after each of the 15 training epochs.

Final test accuracy is **83.87%**. The highest observed test accuracy is 86.13% at epoch 11. Training loss continues falling, from 0.0650 at epoch 11 to 0.0198 at epoch 15, while test accuracy falls by 2.27 percentage points. This suggests overfitting and generalization fluctuations on a small dataset. I retain the required 15-epoch result rather than selecting a stopping point using the test set.

Question 1.4: Sanity Checks – Attention Figures and Discussion

I checked that every attention row sums to one across every head and layer for the two example sentences. The maximum row-sum errors were 1.79×10^{-7} and 2.38×10^{-7}, respectively. Both checks passed. Figures 2 and 3 show layer 4, head 1; rows are queries and columns are attended-to keys.

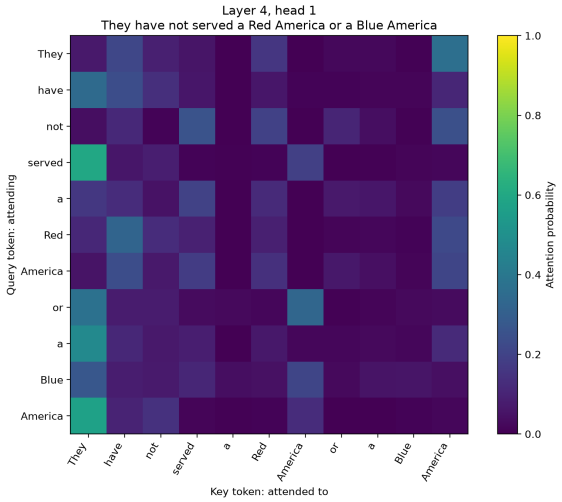


Figure 2: Encoder attention for the Obama sentence; layer 4, head 1.

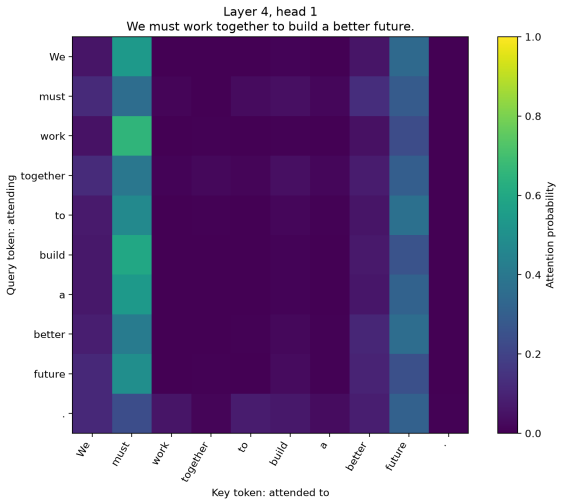


Figure 3: Encoder attention for the second sentence; layer 4, head 1.

The encoder has no causal restriction: attention can go to earlier and later words. In Figure 2, the initial query "They" attends to the final "America," demonstrating access to future context. In Figure 3, many rows emphasize the "must" and "future" columns. This head therefore uses some shared contextual anchors rather than distributing attention uniformly. These are descriptive patterns in one head, not proof that particular attention weights cause a speaker prediction.

Part 2: Pretraining a Decoder Language Model (20 points)

2.1 Decoder. The four-block word-level Transformer decoder uses causal self-attention and a 100-unit ReLU feedforward hidden layer.

2.2 Pretraining. We predict the next word by minimizing cross-entropy for 500 optimizer steps (batch size 16, block size 32).

2.3 Sanity checks. Attention-row normalization and strict causality are tested; attention maps and their interpretation are on the next page.

2.4 Evaluation. Perplexities on the three politician test sets are reported every 100 iterations and at the end.

The 500 updates process 256,000 target-token positions; overlapping training windows mean these are not all unique corpus tokens.

Decoder backbone: 482,192 parameters. **Vocabulary projection:** 366,210 parameters. **Total decoder language model:** 848,402 parameters.

Iteration	Train PPL	Obama PPL	W. Bush PPL	H. W. Bush PPL
100	621.46	700.45	808.06	708.30
200	481.78	606.45	674.12	593.14
300	340.80	490.26	565.18	489.01
400	237.09	422.79	488.62	421.47
500	183.19	391.65	472.84	404.10

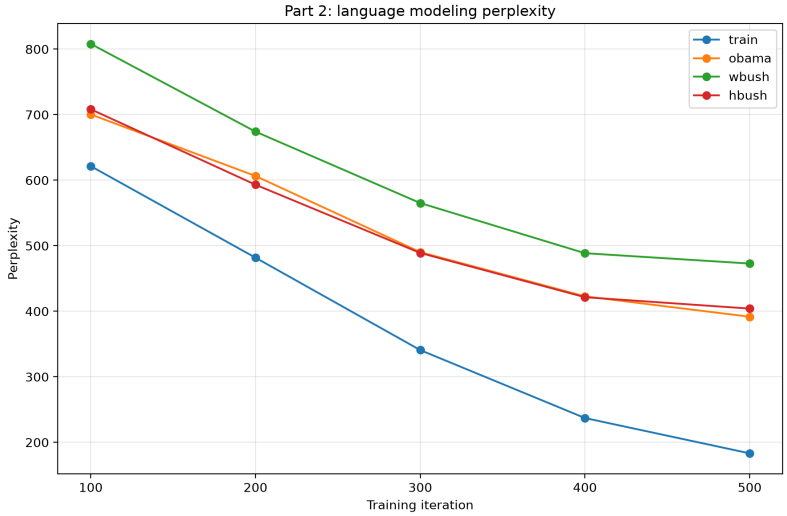


Figure 4: Training and test perplexity, evaluated every 100 updates.

Evaluation protocol and interpretation

Perplexity is exp(mean token cross-entropy). At each evaluation, I use the first 200 batches of the starter-style sliding-window dataset for each corpus. Each evaluated corpus contributes 102,400 target positions, with repeated tokens due to overlapping windows. These are fixed-prefix window estimates, not full-file, unique-token perplexities. The loss is weighted by the number of target tokens before exponentiation.

Training perplexity decreases from 621.46 to 183.19. All three test perplexities also decrease at every checkpoint, reaching 391.65 for Obama, 472.84 for W. Bush, and 404.10 for H. W. Bush. W. Bush is consistently the most difficult evaluated corpus. Differences in wording, style, unknown-token rates, or similarity to training text could contribute, but these aggregate measurements do not identify a specific cause.

The gap between final training and test perplexity indicates better fit to the training text than to these held-out speeches. The test curves are still improving at update 500; I keep the prescribed training budget for comparability.

Question 2.3: Sanity Checks – Causal Attention Figures

Both normalization checks passed, with maximum row-sum errors of 1.79×10^{-7} and 1.19×10^{-7}. The utility also checked that every strictly upper-triangular attention entry is zero. A separate whole-model check changed the last input token and verified that earlier logits were unchanged; it passed before and after training.

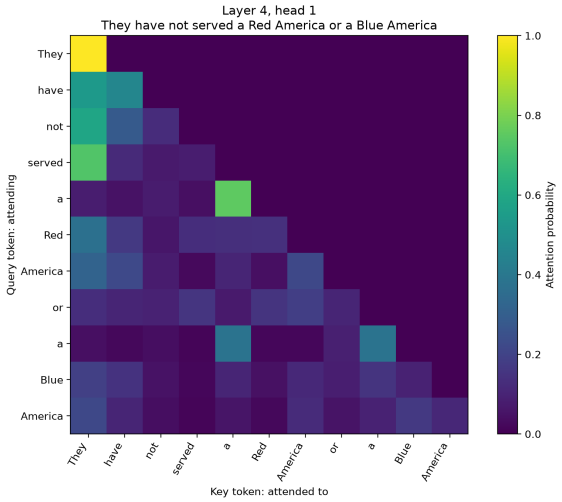


Figure 5: Causal decoder attention for the Obama sentence; layer 4, head 1.

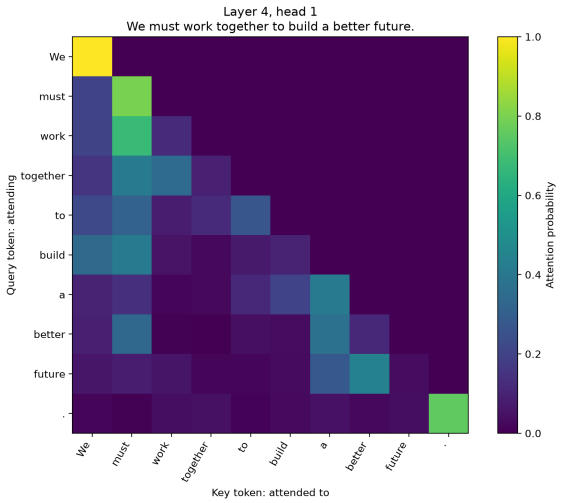


Figure 6: Causal decoder attention for the second sentence; layer 4, head 1.

The upper-right triangle is zero in both figures, showing that each query can attend only to itself and earlier tokens. The first query has only one legal key, so its self-attention probability is one. Later queries distribute probability across an expanding prefix. In Figure 6, the "work" query emphasizes "must," while the period strongly emphasizes itself. Figure 5 also shows attention to earlier occurrences of "a." These patterns respect causal access, unlike the bidirectional encoder plots.

Together, the mask checks, future-token invariance check, and decreasing held-out perplexities support a functioning causal language model. Normalized attention alone would not demonstrate that the mask was correct; the additional causal checks address that distinction.

CSE 256 PA2 — Part B: Executed Part 3 Notebook

Code cell [5]

```
import torch
import transformers
print("Name: Jai Dayanand")
print("torch:", torch.__version__)
print("transformers:", transformers.__version__)
```

```
Name: Jai Dayanand
torch: 2.15.0+cpu
transformers: 5.22.2
```

CSE 256 PA2 — Part 3: Dissecting Pythia

Jai Dayanand

The following analyses use `EleutherAI/pythia-160m-deduped` at checkpoints throughout pretraining. Each experiment reports its numerical outputs, figures, and interpretations alongside the code that produced them. Evaluation uses the pretrained Pythia tokenizer and float32 model weights. Reproducing the experiments requires the speech test data and access to the checkpoint weights.

Code cell [6]

```
from pathlib import Path
import gc, json, hashlib, platform
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.colors import LogNorm
from IPython.display import display
from transformers import AutoTokenizer, GPTNeoXForCausalLM

SEED = 42
torch.manual_seed(SEED)
np.random.seed(SEED)
DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
MODEL_NAME = "EleutherAI/pythia-160m-deduped"
FINAL = "step143000"
CONTEXT = 128
READOUT_CHUNK = 32

# Find the data whether Jupyter starts in PA2_code or its parent.
candidates = [Path("speechesdataset/test_LM_obama.txt"),
              Path("PA2_code/speechesdataset/test_LM_obama.txt")]
DATA_PATH = next((p for p in candidates if p.is_file()), None)
if DATA_PATH is None:
    raise FileNotFoundError("Open this notebook from PA2_code, or edit DATA_PATH.")
OUT = DATA_PATH.parent.parent / "outputs" / "part3"
OUT.mkdir(parents=True, exist_ok=True)
text = DATA_PATH.read_text(encoding="utf-8")
tok = AutoTokenizer.from_pretrained(MODEL_NAME, revision=FINAL)
all_ids = tok(text, add_special_tokens=False, return_tensors="pt",
              verbose=False).input_ids[0]
assert len(all_ids) > 1
print("Device:", DEVICE, "| Python:", platform.python_version())
print("Data:", DATA_PATH.resolve())
print("Pythia tokens:", len(all_ids), "| predicted transitions:", len(all_ids)-1)
print("Output folder:", OUT.resolve())
print("Evaluation: whole file; 128 prediction positions per window;")
print("adjacent windows share one boundary token; context resets each window.")

manifest = dict(model=MODEL_NAME, context=CONTEXT, seed=SEED,
                torch=str(torch.__version__), transformers=transformers.__version__,
                data_sha256=hashlib.sha256(DATA_PATH.read_bytes()).hexdigest(),
                token_count=len(all_ids), dtype="float32")
(OUT / "run_config.json").write_text(json.dumps(manifest, indent=2))
```

```
Device: cpu | Python: 3.12.7
Data: C:\Users\Jai
```

```
Dayanand\Downloads\UCSD_Assignments\CSE256\CSE256_PA2_FA26\PA2_code\speechesdataset\test_LM_obama.txt
Pythia tokens: 6017 | predicted transitions: 6016
Output folder: C:\Users\Jai Dayanand\Downloads\UCSD_Assignments\CSE256\CSE256_PA2_FA26\PA2_code\outputs\part3
Evaluation: whole file; 128 prediction positions per window;
adjacent windows share one boundary token; context resets each window.

262
```

3.1 Reading Pythia

Read Biderman et al. (2023), especially Sections 2.6 and 3.1--3.3.

3.1 — Question 1: Why checkpoints? (2 points)

The **memorization study (Section 3.2)** depends most on Pythia's checkpoint and fixed-data-order design: the authors examine the occurrence of memorized sequences over training, using the known training-data sequence to study their dependence on location and time. For a study that could **mostly** be carried out with final models, I would choose a *static version* of the **term-frequency/few-shot-accuracy study (Section 3.3)**: one could compare final-model task performance against pretraining term frequencies. However, doing so would **not** reproduce the paper's central temporal finding, namely that the relationship emerges partway through pretraining. The gender-bias intervention (Section 3.1) also requires access to earlier training states to resume training on modified data, so the complete experiment cannot be reduced to only final-model evaluation.

3.1 — Question 2: Deduplicated versus original data (2 points)

Section 2.6 reports no clear language-modeling performance benefit from deduplication in this controlled suite. This is a finding about these experiments, rather than a universal claim that deduplication is ineffective.

Reference: Biderman et al., *Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling*, ICML 2023, Sections 2.6 and 3.2–3.3.

Shared analysis functions

The logit lens applies the checkpoint's own final layer norm and output projection to intermediate residual states. The final returned hidden state is checked against the actual logits to avoid applying final layer norm twice. Layer 0 is explicitly read from the embedding weights.

The evaluation predicts **every adjacent token pair in the full file exactly once**. At window boundaries context resets. All checkpoints and transplants use this identical protocol; these subword-token metrics are not directly comparable with Part 2's word-level perplexities.

Code cell [7]

```
def release_memory():
    gc.collect()
    if torch.cuda.is_available():
        torch.cuda.empty_cache()


def load_model(revision):
    print("Loading", revision, flush=True)
    m = GPTNeoXForCausalLM.from_pretrained(
        MODEL_NAME, revision=revision,
        torch_dtype=torch.float32, attn_implementation="eager",
    ).to(DEVICE).eval()
    m.requires_grad_(False)
    assert m.config.num_hidden_layers == 12
    assert m.config.hidden_size == 768
    assert m.config.num_attention_heads == 12
    assert next(m.parameters()).dtype == torch.float32
    return m


def output_head(m):
    return m.get_output_embeddings()


def windows():
```

```
    for start in range(0, len(all_ids)-1, CONTEXT):
        # Every window has one extra token for shifted targets.
        w = all_ids[start:start+CONTEXT+1].unsqueeze(0).to(DEVICE)
        yield w


@torch.inference_mode()
def verify_hidden_convention(m):
    x = all_ids[:12].unsqueeze(0).to(DEVICE)
    out = m(x, output_hidden_states=True, use_cache=False)
    assert len(out.hidden_states) == 13
    direct = output_head(m)(out.hidden_states[-1])
    normalized = output_head(m)(m.gpt_neox.final_layer_norm(out.hidden_states[-1]))
    direct_error = (direct-out.logits).abs().max().item()
    normalized_error = (normalized-out.logits).abs().max().item()
    already_normalized = torch.allclose(direct, out.logits, atol=1e-5, rtol=1e-5)
    if not already_normalized:
        assert torch.allclose(normalized, out.logits, atol=1e-5, rtol=1e-5)
    embedded = m.get_input_embeddings()(x)
    assert torch.allclose(embedded, out.hidden_states[0], atol=1e-5, rtol=1e-5)
    print("Final hidden state already normalized:", already_normalized)
    print("Logit max errors — direct:", direct_error,
          "after another LN:", normalized_error)
    return already_normalized


def lens_logits(m, hidden, layer, already_normalized):
    if layer == 12 and already_normalized:
        return output_head(m)(hidden)
    return output_head(m)(m.gpt_neox.final_layer_norm(hidden))


@torch.inference_mode()
def lens_accuracy(m, already_normalized):
    correct = np.zeros(13, dtype=np.int64)
    count = 0
    for w in windows():
        out = m(w[:, :-1], output_hidden_states=True, use_cache=False)
        targets = w[:, 1:]
        for layer, hidden in enumerate(out.hidden_states):
            if layer == 0:
                hidden = m.get_input_embeddings()(w[:, :-1])
            for start in range(0, targets.shape[1], READOUT_CHUNK):
                stop = start + READOUT_CHUNK
                logits = lens_logits(m, hidden[:, start:stop], layer,
                                     already_normalized)
                correct[layer] += (logits.argmax(-1) == targets[:, start:stop]).sum().item()
        count += targets.numel()
    assert count == len(all_ids)-1
    return correct / count, correct, count


@torch.inference_mode()
def corpus_loss(m):
    total, count = 0.0, 0
    for w in windows():
        logits = m(w[:, :-1], use_cache=False).logits
        targets = w[:, 1:]
        for start in range(0, targets.shape[1], READOUT_CHUNK):
            stop = start + READOUT_CHUNK
            total += torch.nn.functional.cross_entropy(
                logits[:, start:stop].reshape(-1, logits.shape[-1]),
                targets[:, start:stop].reshape(-1), reduction="sum").item()
        count += targets.numel()
    assert count == len(all_ids)-1
    return total/count


def token_label(token_id):
    return tok.decode([int(token_id)]).replace(" ", "■").replace("\n", "\\n")


@torch.inference_mode()
def lens_grid(m, sentence, already_normalized, filename):
    ids = tok(sentence, add_special_tokens=False, return_tensors="pt").input_ids.to(DEVICE)
    assert 2 <= ids.shape[1] <= m.config.max_position_embeddings
    out = m(ids[:, :-1], output_hidden_states=True, use_cache=False)
```

```
    targets = ids[:, 1:]
    probabilities, predictions = [], []
    for layer, hidden in enumerate(out.hidden_states):
        if layer == 0:
            hidden = m.get_input_embeddings()(ids[:, :-1])
        logits = lens_logits(m, hidden, layer, already_normalized)
        probs = logits.softmax(-1)
        probabilities.append(probs.gather(-1, targets.unsqueeze(-1)).squeeze().reshape(-1).cpu().numpy())
        predictions.append(logits.argmax(-1)[0].cpu().numpy())
    probabilities = np.stack(probabilities)
    predictions = np.stack(predictions)
    fig, ax = plt.subplots(figsize=(max(10, targets.numel()*0.8), 8))
    im = ax.imshow(probabilities, origin="lower", aspect="auto", vmin=0, vmax=1, cmap="Blues")
    for layer in range(13):
        for pos in range(targets.numel()):
            ax.text(pos, layer, token_label(predictions[layer, pos]),
                    ha="center", va="center", fontsize=7,
                    color="white" if probabilities[layer, pos] > .5 else "black")
    ax.set_xticks(range(targets.numel()))
    ax.set_xticklabels([token_label(i) for i in ids[0, :-1]], rotation=60, ha="right")
    ax.set_yticks(range(13))
    ax.set_ylabel("Layer (0 = embedding)")
    ax.set_xlabel("Input token; each cell predicts the following token")
    ax.set_title(sentence + "\nCell text: top prediction; shade: true next-token probability")
    fig.colorbar(im, ax=ax, label="Probability of true next token")
    fig.tight_layout(); fig.savefig(OUT / filename, dpi=180, bbox_inches="tight")
    plt.show()
    print("Subword tokens:", ids.numel())
    print("True next tokens:", [token_label(i) for i in targets[0]])
    np.savez(OUT / filename.replace(".png", ".npz"), probabilities=probabilities,
             predictions=predictions, ids=ids.cpu().numpy())
```

3.2.1 Watch a prediction form (13 points)

Final checkpoint: `step143000`. Each grid excludes the last input position because the supplied sentence provides no next-token target there.

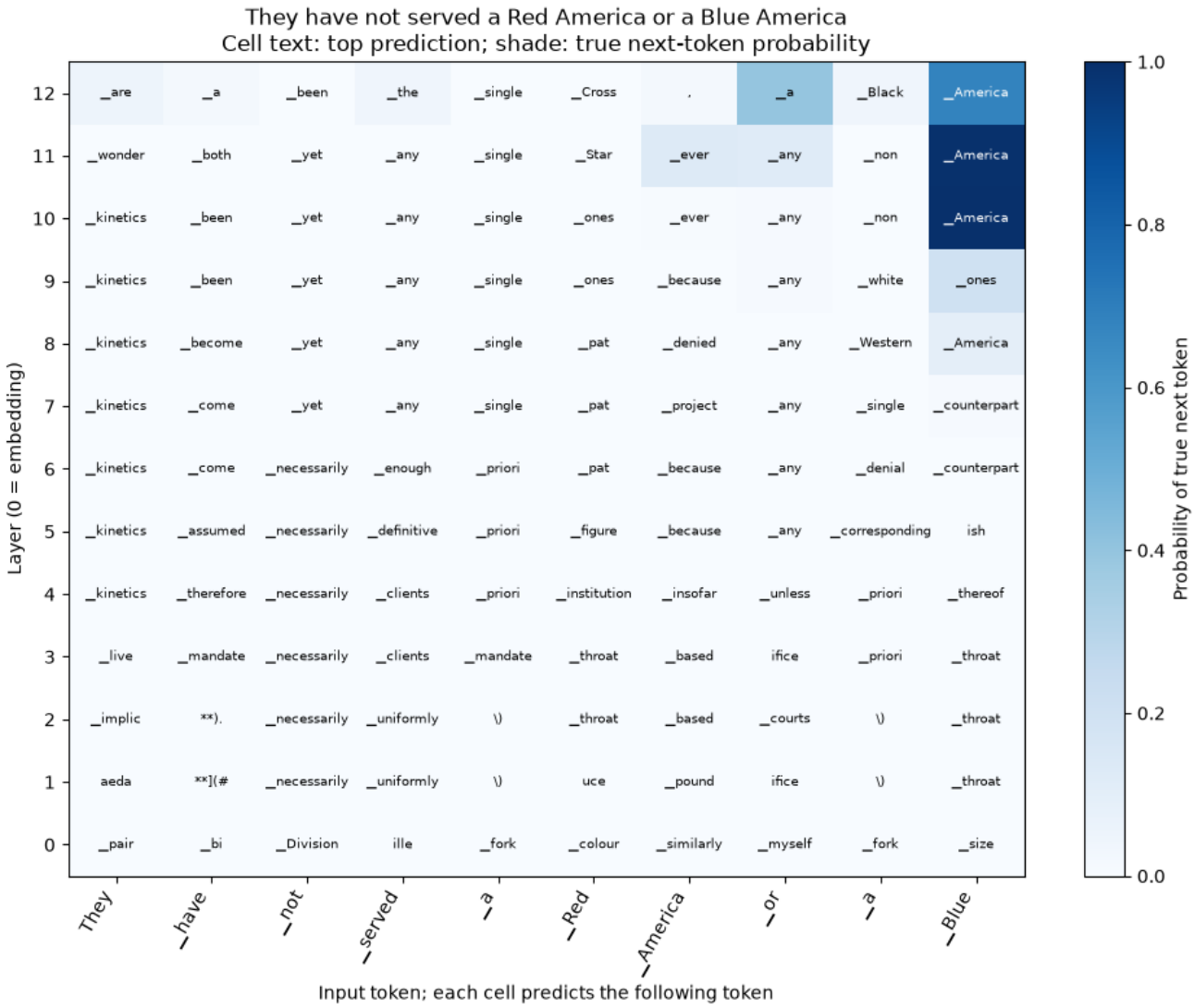
Code cell [8]

```
model = load_model(FINAL)
final_normalized = verify_hidden_convention(model)
sentence_obama = "They have not served a Red America or a Blue America"
# Original sentence used for logit-lens analysis.
sentence_own = "The curious student opened the book and found a surprising answer."
lens_grid(model, sentence_obama, final_normalized, "grid_obama.png")
lens_grid(model, sentence_own, final_normalized, "grid_original.png")
```

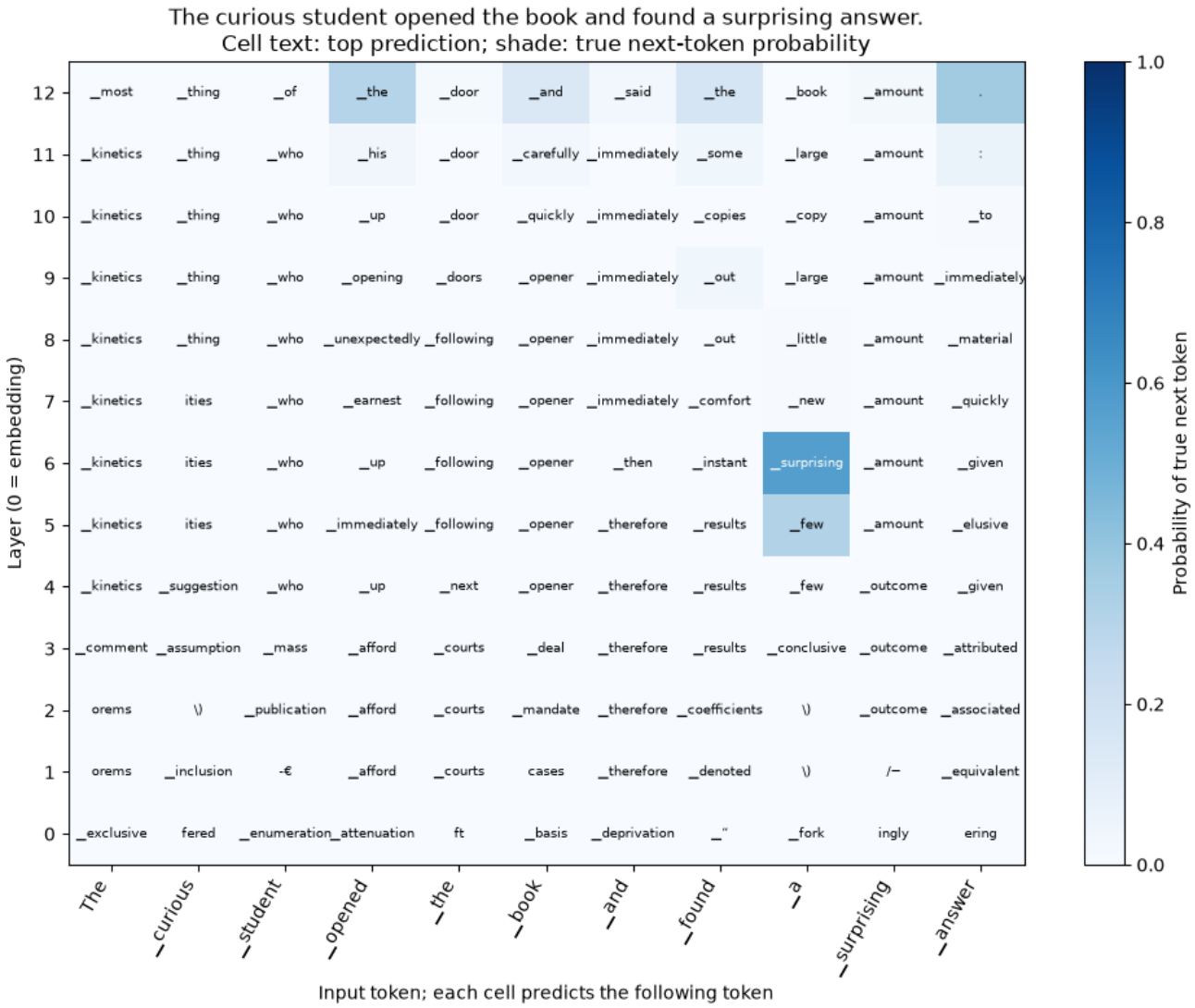
```
Loading step143000
```

```
Loading weights: 100%|■■■■■■■■■■| 148/148 [00:00<00:00, 574.48it/s]
```

```
Final hidden state already normalized: True
Logit max errors — direct: 0.0 after another LN: 7.51715087890625
```




```
Subword tokens: 11
True next tokens: ['■have', '■not', '■served', '■a', '■Red', '■America', '■or', '■a', '■Blue', '■America']
```

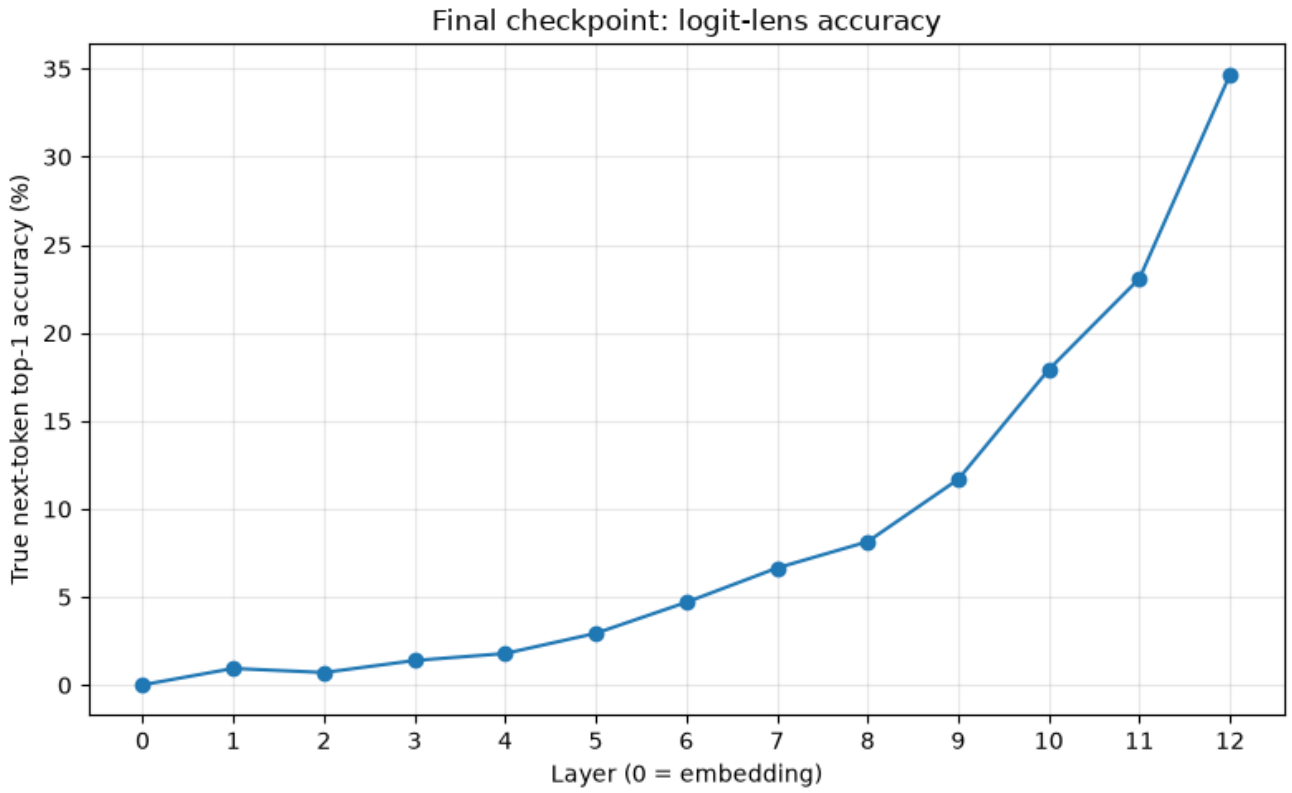



```
Subword tokens: 12
True next tokens: ['■curious', '■student', '■opened', '■the', '■book', '■and', '■found', '■a', '■surprising',
'■answer', '.']
```

Code cell [9]

```
final_accuracy, final_correct, denominator = lens_accuracy(model, final_normalized)
for layer in [0, 6, 12]:
    print(f"Layer {layer}: {final_correct[layer]}/{denominator} = "
          f"{final_accuracy[layer]:.10f} ({100*final_accuracy[layer]:.6f}%)")
fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(range(13), 100*final_accuracy, marker="o")
ax.set(xticks=range(13), xlabel="Layer (0 = embedding)",
       ylabel="True next-token top-1 accuracy (%)", title="Final checkpoint: logit-lens accuracy")
ax.grid(alpha=.3); fig.tight_layout()
fig.savefig(OUT / "final_accuracy.png", dpi=180); plt.show()
pd.DataFrame({"layer": range(13), "correct": final_correct,
              "count": denominator, "accuracy": final_accuracy}).to_csv(OUT / "final_accuracy.csv", index=False)
del model
release_memory()
```

```
Layer 0: 1/6016 = 0.0001662234 (0.016622%)
Layer 6: 283/6016 = 0.0470412234 (4.704122%)
Layer 12: 2086/6016 = 0.3467420213 (34.674202%)
```



3.2.1 Interpretation — final-checkpoint logit lens

The two layer-by-position grids above show the top next-token predictions for the specified Obama passage and for the original sentence, *"The curious student opened the book and found a surprising answer."* Each cell's color reflects the probability assigned to the true next token, not the probability assigned to the displayed top prediction. The grid makes it possible to compare the embedding-only predictions with intermediate and final predictions at each position.

Across all 6,016 next-token pairs in `test_LM_obama.txt`, the embedding-only readout (layer 0) gets **1/6,016 = 0.016622%** correct, layer 6 gets **283/6,016 = 4.704122%**, and the final layer 12 gets **2,086/6,016 = 34.674202%**. This demonstrates substantially better predictive accuracy at the end of the network than at the embedding or midpoint. Improvement is **not strictly monotonic**: accuracy falls from 0.9475% at layer 1 to 0.7148% at layer 2, then rises through the later layers. These are empirical properties of the plotted readouts rather than evidence that each layer independently performs next-word classification.

3.2.2 Across training (13 points)

The same full-file token pairs and context boundaries are used at all four checkpoints. Each checkpoint uses its own final layer norm and unembedding.

Code cell [10]

```
curves = {FINAL: final_accuracy.copy()}
for revision in ["step512", "step1000", "step8000"]:
    model = load_model(revision)
    convention = verify_hidden_convention(model)
    accuracy, correct, count = lens_accuracy(model, convention)
    curves[revision] = accuracy
    print(revision, "layer 0/6/12:", accuracy[[0, 6, 12]])
    del model
    release_memory()

curve_table = pd.DataFrame({"layer": range(13), **curves})
display(curve_table)
curve_table.to_csv(OUT / "checkpoint_accuracy.csv", index=False)
fig, ax = plt.subplots(figsize=(9, 6))
for revision in ["step512", "step1000", "step8000", FINAL]:
```

```
    ax.plot(range(13), 100*curves[revision], marker="o", label=revision)
ax.set(xticks=range(13), xlabel="Layer (0 = embedding)",
       ylabel="True next-token top-1 accuracy (%)", title="Logit-lens readability across pretraining")
ax.legend(); ax.grid(alpha=.3); fig.tight_layout()
fig.savefig(OUT / "checkpoint_accuracy.png", dpi=180); plt.show()
```

```
Loading step512
```

```
Loading weights: 100%|■■■■■■■■■■■| 148/148 [00:00<00:00, 456.70it/s]
```

```
Final hidden state already normalized: True
Logit max errors — direct: 0.0 after another LN: 0.35659098625183105
step512 layer 0/6/12: [0.00049867 0.11486037 0.20329122]
Loading step1000
```

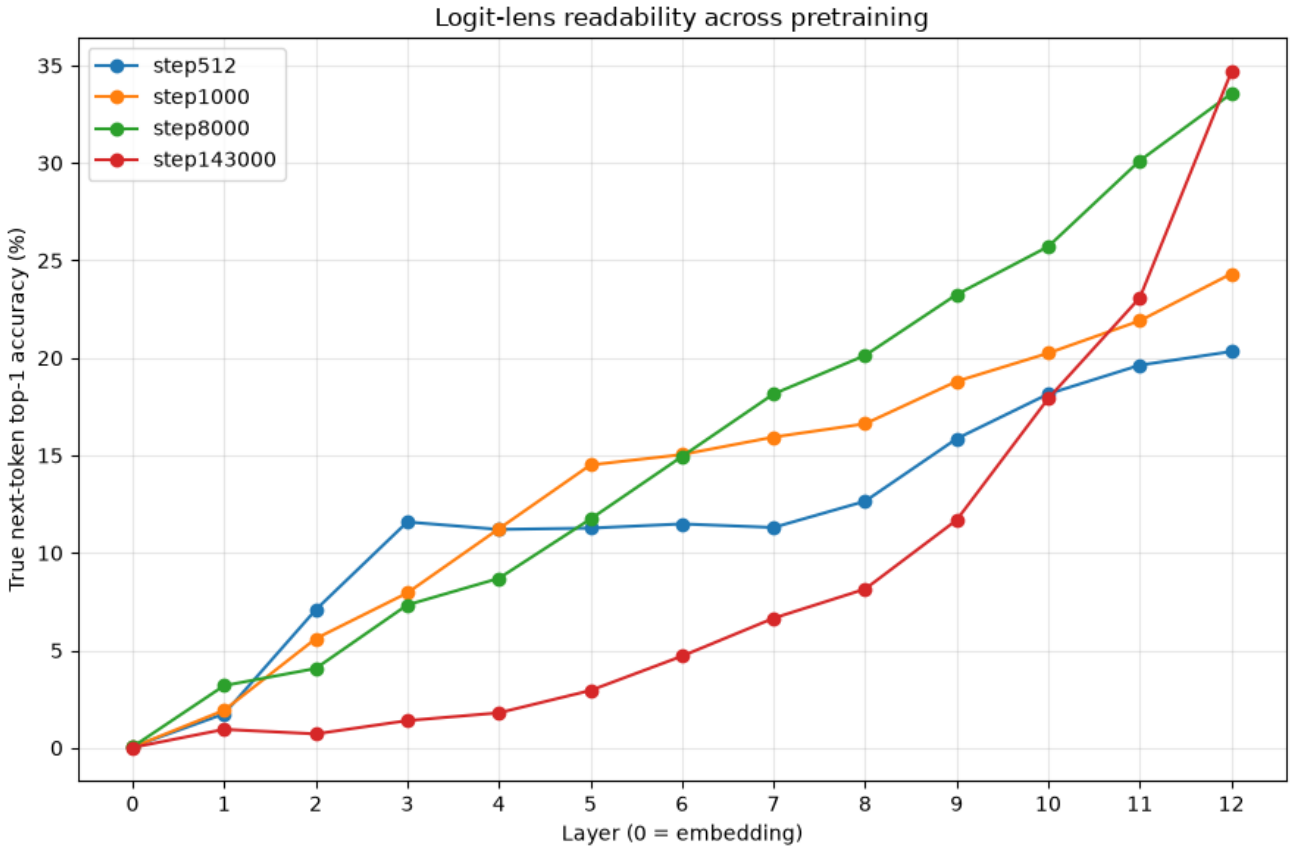
```
Loading weights: 100%|■■■■■■■■■■■| 148/148 [00:00<00:00, 440.27it/s]
```

```
Final hidden state already normalized: True
Logit max errors — direct: 0.0 after another LN: 0.806854248046875
step1000 layer 0/6/12: [0.00033245 0.15043218 0.24318484]
Loading step8000
```

```
Loading weights: 100%|■■■■■■■■■■■| 148/148 [00:00<00:00, 502.62it/s]
```

```
Final hidden state already normalized: True
Logit max errors — direct: 0.0 after another LN: 5.333597183227539
step8000 layer 0/6/12: [0.00049867 0.14943484 0.33560505]
```

	layer	step143000	step512	step1000	step8000
0	0	0.000166	0.000499	0.000332	0.000499
1	1	0.009475	0.017453	0.019282	0.031915
2	2	0.007148	0.070811	0.056017	0.040725
3	3	0.013963	0.115858	0.079455	0.073305
4	4	0.017952	0.112035	0.112367	0.086935
5	5	0.029422	0.112699	0.145113	0.117354
6	6	0.047041	0.114860	0.150432	0.149435
7	7	0.066489	0.113032	0.159408	0.181516
8	8	0.081449	0.126496	0.166223	0.201297
9	9	0.116855	0.158577	0.187999	0.232547
10	10	0.179189	0.181516	0.202460	0.257148
11	11	0.230884	0.196310	0.219082	0.301197
12	12	0.346742	0.203291	0.243185	0.335605



3.2.2 Interpretation — changes across pretraining

The later checkpoint has a substantially better **final** next-token readout: layer-12 accuracy increases from **20.3291%** at step512 to **24.3185%** at step1000, **33.5605%** at step8000, and **34.6742%** at step143000. Intermediate-layer *untuned logit-lens* readability, however, does not improve monotonically as training progresses. At layer 6 the accuracies are **11.4860%**, **15.0432%**, **14.9435%**, and only **4.7041%**, respectively. At layer 8 they are **12.6496%**, **16.6223%**, **20.1297%**, and **8.1449%**. Thus the final model predicts better at the output while its intermediate states are harder to decode using the *final-layer readout*.

One plausible interpretation is that intermediate representations increasingly differ from the coordinate system or feature distribution expected by the checkpoint's final normalization and unembedding. A raw logit lens applies this final readout to states it was not trained to interpret directly. Consequently, low middle-layer lens accuracy **does not establish that next-token information has been lost**. Layer-specific readouts such as the tuned lens are designed to separate readout misalignment from information that is genuinely unavailable. The differences across these checkpoints are not uniform: for example, step1000 is slightly better than step8000 at layer 6, while step8000 is better at layer 8.

Optional context: Belrose et al. (2023), *Eliciting Latent Predictions from Transformers with the Tuned Lens*.

3.3 Transplanting layers across time (20 points)

Recipient: intact `step143000`. Donors: `step0`, `step1000`, `step8000`, `step32000`.

Components: token embedding, each of the 12 complete Transformer blocks, and output unembedding. The recipient's final layer norm is held fixed. Blocks are numbered **1–12** here (Hugging Face indices 0–11). The embedding and unembedding are separate, untied components. The metric is the change in mean next-token cross-entropy, in nats/token.

Each transplant is restored in a `finally` block and checked tensor by tensor. The baseline is rechecked after each donor. The full corpus is evaluated for every pair.

Code cell [11]

```
def components(m):
    return {"embedding": m.get_input_embeddings(),
```

```
            **{f"block_{i+1:02d}": block for i, block in enumerate(m.gpt_neox.layers)},
            "unembedding": output_head(m)}


def snapshot(module):
    return {key: value.detach().cpu().clone()
            for key, value in module.state_dict().items()}


model = load_model(FINAL)
assert model.get_input_embeddings().weight.data_ptr() != output_head(model).weight.data_ptr(), \
    "This experiment assumes untied input/output weights."
baseline = corpus_loss(model)
print(f"Intact final model baseline: {baseline:.10f} nats/token")
recipient_components = components(model)
component_names = list(recipient_components)
donors = ["step0", "step1000", "step8000", "step32000"]
deltas = pd.DataFrame(index=component_names, columns=donors, dtype=float)
losses = deltas.copy()

for revision in donors:
    donor = load_model(revision)
    donor_components = components(donor)
    for name, recipient_module in recipient_components.items():
        original = snapshot(recipient_module)
        try:
            recipient_module.load_state_dict(donor_components[name].state_dict(), strict=True)
            transplanted_loss = corpus_loss(model)
            losses.loc[name, revision] = transplanted_loss
            deltas.loc[name, revision] = transplanted_loss - baseline
            print(f"{revision:>10} | {name:>12} | loss {transplanted_loss:.8f} | "
                  f"delta {transplanted_loss-baseline:+.8f}", flush=True)
        finally:
            recipient_module.load_state_dict(original, strict=True)
            for key, value in recipient_module.state_dict().items():
                assert torch.equal(value.detach().cpu(), original[key]), f"Restore failed: {name}/{key}"
        del original
        # Save progress, but reruns recompute rather than silently reusing old cells.
        deltas.to_csv(OUT / "transplant_deltas.csv")
        losses.to_csv(OUT / "transplant_losses.csv")
    restored_loss = corpus_loss(model)
    assert abs(restored_loss-baseline) < 1e-5, "Recipient baseline changed."
    print("Restored baseline check PASSED:", restored_loss)
    del donor_components, donor
    release_memory()

print("Baseline loss (nats/token):", baseline)
display(deltas)
(OUT / "baseline.json").write_text(json.dumps({"loss_nats_per_token": baseline}, indent=2))

Loading step143000

Loading weights: 100%|■■■■■■■■■■| 148/148 [00:00<00:00, 567.64it/s]

Intact final model baseline: 3.4617656152 nats/token
Loading step0

Loading weights: 100%|■■■■■■■■■■| 148/148 [00:00<00:00, 1112.27it/s]

     step0 |    embedding | loss 8.57390565 | delta +5.11214004
     step0 |     block_01 | loss 8.65458387 | delta +5.19281825
     step0 |     block_02 | loss 4.23498340 | delta +0.77321778
     step0 |     block_03 | loss 4.46389110 | delta +1.00212548
     step0 |     block_04 | loss 4.31632560 | delta +0.85455998
     step0 |     block_05 | loss 3.94973858 | delta +0.48797296
     step0 |     block_06 | loss 3.88774278 | delta +0.42597717
     step0 |     block_07 | loss 3.82442032 | delta +0.36265470
     step0 |     block_08 | loss 3.78909697 | delta +0.32733136
     step0 |     block_09 | loss 3.83226879 | delta +0.37050318
     step0 |     block_10 | loss 3.90834231 | delta +0.44657670
     step0 |     block_11 | loss 4.24886684 | delta +0.78710123
     step0 |     block_12 | loss 8.32575322 | delta +4.86398760
     step0 |  unembedding | loss 20.03662766 | delta +16.57486204
Restored baseline check PASSED: 3.4617656152299108
Loading step1000

Loading weights: 100%|■■■■■■■■■■| 148/148 [00:00<00:00, 880.98it/s]

  step1000 |    embedding | loss 6.13314696 | delta +2.67138134
```

```
  step1000 |     block_01 | loss 8.35337168 | delta +4.89160606
  step1000 |     block_02 | loss 4.24538796 | delta +0.78362235
  step1000 |     block_03 | loss 4.33661107 | delta +0.87484546
  step1000 |     block_04 | loss 4.37034449 | delta +0.90857888
  step1000 |     block_05 | loss 3.98832057 | delta +0.52655496
  step1000 |     block_06 | loss 3.92483660 | delta +0.46307098
  step1000 |     block_07 | loss 3.78534498 | delta +0.32357937
  step1000 |     block_08 | loss 3.78051803 | delta +0.31875241
  step1000 |     block_09 | loss 3.83833280 | delta +0.37656718
  step1000 |     block_10 | loss 3.87554246 | delta +0.41377684
  step1000 |     block_11 | loss 4.25474393 | delta +0.79297832
  step1000 |     block_12 | loss 8.70758731 | delta +5.24582169
  step1000 |  unembedding | loss 33.23892708 | delta +29.77716147
Restored baseline check PASSED: 3.4617656152299108
Loading step8000

Loading weights: 100%|■■■■■■■■■■| 148/148 [00:00<00:00, 448.88it/s]

  step8000 |    embedding | loss 4.02415370 | delta +0.56238809
  step8000 |     block_01 | loss 4.53971046 | delta +1.07794485
  step8000 |     block_02 | loss 4.47155310 | delta +1.00978749
  step8000 |     block_03 | loss 4.30001637 | delta +0.83825075
  step8000 |     block_04 | loss 4.91302312 | delta +1.45125751
  step8000 |     block_05 | loss 6.71922260 | delta +3.25745698
  step8000 |     block_06 | loss 4.36387645 | delta +0.90211084
  step8000 |     block_07 | loss 3.99850225 | delta +0.53673663
  step8000 |     block_08 | loss 4.27145180 | delta +0.80968618
  step8000 |     block_09 | loss 4.25984527 | delta +0.79807966
  step8000 |     block_10 | loss 4.45662183 | delta +0.99485622
  step8000 |     block_11 | loss 4.59574738 | delta +1.13398177
  step8000 |     block_12 | loss 9.13198382 | delta +5.67021820
  step8000 |  unembedding | loss 52.03067228 | delta +48.56890666
Restored baseline check PASSED: 3.4617656152299108
Loading step32000

Loading weights: 100%|■■■■■■■■■■| 148/148 [00:00<00:00, 1173.78it/s]

 step32000 |    embedding | loss 3.60539660 | delta +0.14363099
 step32000 |     block_01 | loss 4.39039314 | delta +0.92862753
 step32000 |     block_02 | loss 4.26604455 | delta +0.80427893
 step32000 |     block_03 | loss 4.07385172 | delta +0.61208611
 step32000 |     block_04 | loss 4.38346221 | delta +0.92169660
 step32000 |     block_05 | loss 4.05550865 | delta +0.59374303
 step32000 |     block_06 | loss 3.95176445 | delta +0.48999883
 step32000 |     block_07 | loss 3.90958794 | delta +0.44782232
 step32000 |     block_08 | loss 4.35557126 | delta +0.89380565
 step32000 |     block_09 | loss 4.13749046 | delta +0.67572484
 step32000 |     block_10 | loss 4.55420382 | delta +1.09243820
 step32000 |     block_11 | loss 5.35267118 | delta +1.89090556
 step32000 |     block_12 | loss 8.94665640 | delta +5.48489078
 step32000 |  unembedding | loss 83.96371306 | delta +80.50194744
Restored baseline check PASSED: 3.4617656152299108
Baseline loss (nats/token): 3.4617656152299108

                step0   step1000   step8000  step32000
embedding    5.112140   2.671381   0.562388   0.143631
block_01     5.192818   4.891606   1.077945   0.928628
block_02     0.773218   0.783622   1.009787   0.804279
block_03     1.002125   0.874845   0.838251   0.612086
block_04     0.854560   0.908579   1.451258   0.921697
block_05     0.487973   0.526555   3.257457   0.593743
block_06     0.425977   0.463071   0.902111   0.489999
block_07     0.362655   0.323579   0.536737   0.447822
block_08     0.327331   0.318752   0.809686   0.893806
block_09     0.370503   0.376567   0.798080   0.675725
block_10     0.446577   0.413777   0.994856   1.092438
block_11     0.787101   0.792978   1.133982   1.890906
block_12     4.863988   5.245822   5.670218   5.484891
unembedding 16.574862  29.777161  48.568907  80.501947

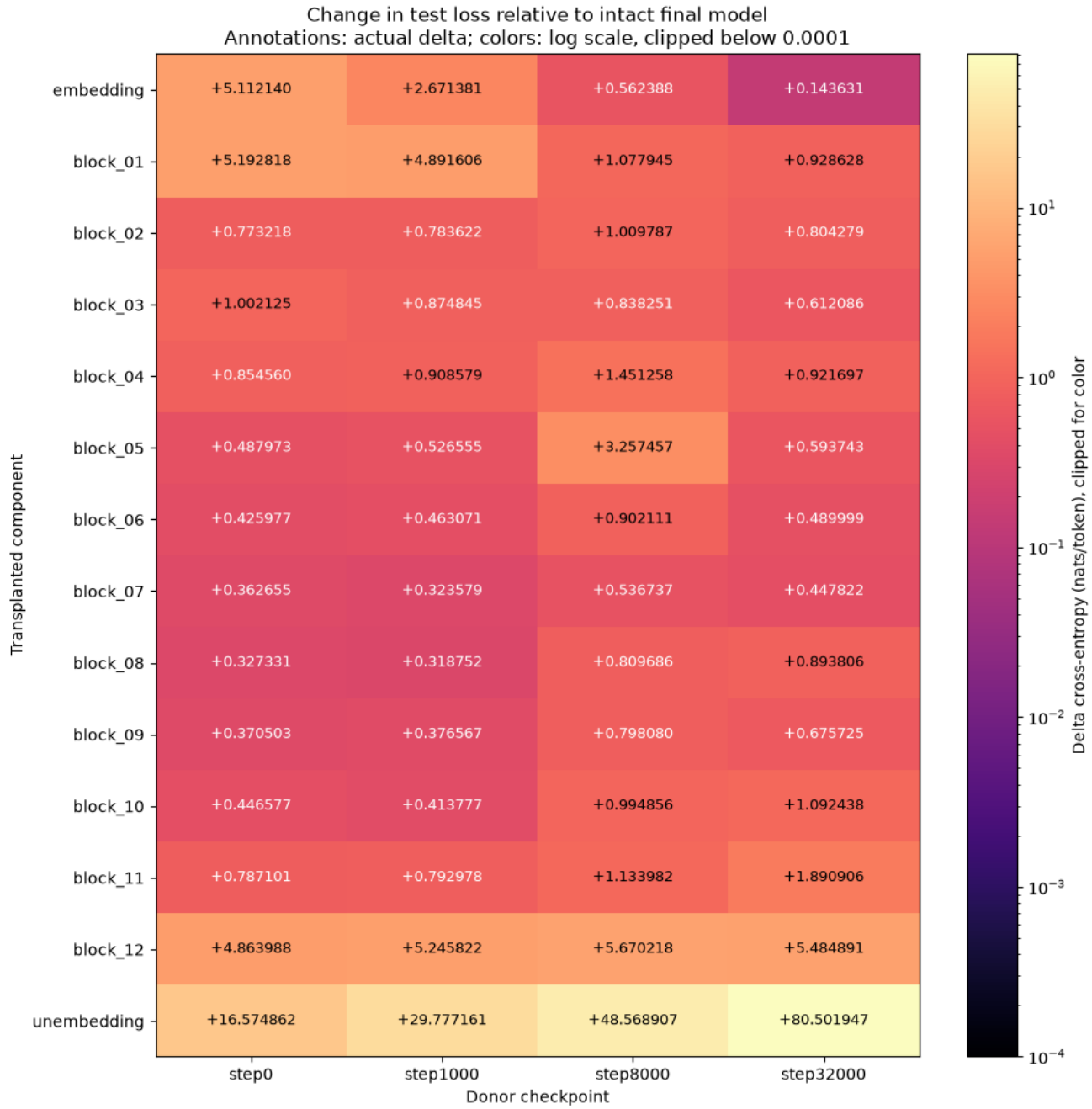
47
```

Code cell [12]

```
values = deltas.to_numpy(dtype=float)
assert np.isfinite(values).all()
floor = 1e-4
ceiling = max(float(values.max()), floor*10)
```

```
# Only the displayed colors are clipped; annotations and CSV retain real deltas.
colors = np.clip(values, floor, ceiling)
fig, ax = plt.subplots(figsize=(10, 10))
im = ax.imshow(colors, aspect="auto", cmap="magma",
               norm=LogNorm(vmin=floor, vmax=ceiling))
ax.set_xticks(range(len(donors)))
ax.set_xticklabels(donors)
ax.set_yticks(range(len(component_names)))
ax.set_yticklabels(component_names)
for row in range(len(component_names)):
    for col in range(len(donors)):
        rgba = im.cmap(im.norm(colors[row, col]))
        luminance = .2126*rgba[0] + .7152*rgba[1] + .0722*rgba[2]
        ax.text(col, row, f"{values[row, col]:+.6f}", ha="center", va="center",
                fontsize=9, color="black" if luminance > .5 else "white")
ax.set_xlabel("Donor checkpoint")
ax.set_ylabel("Transplanted component")
ax.set_title("Change in test loss relative to intact final model\n"
             "Annotations: actual delta; colors: log scale, clipped below 0.0001")
fig.colorbar(im, ax=ax, label="Delta cross-entropy (nats/token), clipped for color")
fig.tight_layout(); fig.savefig(OUT / "transplant_heatmap.png", dpi=180, bbox_inches="tight")
plt.show()

print("Random-donor damage, smallest to largest:")
display(deltas["step0"].sort_values().to_frame("delta_loss"))
print("Adjacent donor pairs where a later donor is more damaging:")
violations = []
for name in component_names:
    for earlier, later in zip(donors[:-1], donors[1:]):
        if deltas.loc[name, later] > deltas.loc[name, earlier] + 1e-6:
            violations.append(dict(component=name, earlier=earlier, later=later,
                                   earlier_delta=deltas.loc[name, earlier],
                                   later_delta=deltas.loc[name, later]))
if violations:
    display(pd.DataFrame(violations))
else:
    print("None under the stated numerical tolerance.")
del recipient_components, model
release_memory()
```



Random-donor damage, smallest to largest:

	delta_loss
block_08	0.327331
block_07	0.362655
block_09	0.370503
block_06	0.425977
block_10	0.446577
block_05	0.487973
block_02	0.773218
block_11	0.787101
block_04	0.854560
block_03	1.002125
block_12	4.863988
embedding	5.112140
block_01	5.192818
unembedding	16.574862

Adjacent donor pairs where a later donor is more damaging:

component	earlier	later	earlier_delta	later_delta

0	block_02	step0	step1000	0.773218	0.783622
1	block_02	step1000	step8000	0.783622	1.009787
2	block_04	step0	step1000	0.854560	0.908579
3	block_04	step1000	step8000	0.908579	1.451258
4	block_05	step0	step1000	0.487973	0.526555
5	block_05	step1000	step8000	0.526555	3.257457
6	block_06	step0	step1000	0.425977	0.463071
7	block_06	step1000	step8000	0.463071	0.902111
8	block_07	step1000	step8000	0.323579	0.536737
9	block_08	step1000	step8000	0.318752	0.809686
10	block_08	step8000	step32000	0.809686	0.893806
11	block_09	step0	step1000	0.370503	0.376567
12	block_09	step1000	step8000	0.376567	0.798080
13	block_10	step1000	step8000	0.413777	0.994856
14	block_10	step8000	step32000	0.994856	1.092438
15	block_11	step0	step1000	0.787101	0.792978
16	block_11	step1000	step8000	0.792978	1.133982
17	block_11	step8000	step32000	1.133982	1.890906
18	block_12	step0	step1000	4.863988	5.245822
19	block_12	step1000	step8000	5.245822	5.670218
20	unembedding	step0	step1000	16.574862	29.777161
21	unembedding	step1000	step8000	29.777161	48.568907
22	unembedding	step8000	step32000	48.568907	80.501947

3.3 Interpretation — which older components can be transplanted?

The intact step143000 recipient has a baseline mean next-token cross-entropy of **3.461766 nats/token**. Replacing one component with step0 weights causes comparatively little damage for several middle blocks: block 8 increases loss by only **0.327331**, block 7 by **0.362655**, and block 9 by **0.370503 nats/token**. By comparison, the **unembedding** is especially difficult to replace (**+16.574862**), followed by block 1 (**+5.192818**), the token embedding (**+5.112140**), and block 12 (**+4.863988**). Here “little damage” is relative to the very large losses observed for the most critical components; even a +0.33 increase is measurable.

A later donor does not always hurt less. The unembedding transplant becomes dramatically worse from step0 (**+16.574862**) to step32000 (**+80.501947**). Block 5 gives another striking counterexample: step1000 causes **+0.526555**, but the later step8000 donor causes **+3.257457**, before the step32000 donor drops to **+0.593743**. Block 12 likewise grows from **+4.863988** at step0 to **+5.670218** at step8000. These numbers rule out a simple monotonic compatibility story. A plausible explanation is that an older component's representations and the fixed final recipient's surrounding components can become *less compatible* at particular intermediate training stages, even while the donor model itself improves. The experiment establishes the incompatibility effects; it does not isolate their mechanistic cause.

Each heatmap cell contains the measured change in corpus test loss, and only the color scale is clipped for visualization. All transplants were evaluated against the same intact final-model baseline, with component weights restored and checked after each replacement.